

Selected SapBERT Model: Final Report

Current four-level dataset | Selected classifier only

The deployed pipeline uses frozen SapBERT, PCA-64 and balanced Logistic Regression (C=100). This report describes the selected model only. It excludes historical baseline tables and other classifier families.

Metric	Grouped CV mean	Retrospective test
Accuracy	90.49%	90.60%
Macro precision	90.75%	90.98%
Macro recall	90.97%	91.00%
Macro F1	90.83%	90.99%

All four aggregate metrics exceed 90% in these evaluations. Individual class scores, shown later, do not all exceed 90%.

Dataset and prediction target

The supplied cardiac_multilingual_10000_4level_triage.xlsx dataset contains 10,000 records. The unchanged partition has 8,001 development records and 1,999 test records. Targets are 0 Emergency, 1 Urgent, 2 Standard and 3 Non-urgent. Supplied labels were not edited to improve scores.

What this report establishes

All figures are plotted from saved experiment data or fitted model artifacts; no decorative or invented clusters are used. The four embedding panels are two-dimensional projections of actual paired-text SapBERT embeddings. A plot shows different evidence from classifier accuracy, as explained on page 5.

These test records were examined previously. Reported test scores are retrospective and require independent confirmation on new records. Label-assignment rules and independent per-record clinical review were not documented. No clinical validation or end-to-end live translation accuracy is claimed.

Selected Pipeline and Input Features

1. Prepare the complaint

During the study, SapBERT receives the supplied/recovered English clinical concept followed by [SEP] and the original complaint. In the GUI, local Ollama supplies the English translation and the anatomical safety check runs before classification. The original complaint is also retained.

2. Generate the embedding

The frozen SapBERT-from-PubMedBERT-fulltext checkpoint uses CLS pooling, L2 normalization and a 128-token input limit. The output is a 768-dimensional vector. SapBERT weights were not fine-tuned.

3. Reduce dimensions

A fitted PCA transform reduces the vector to 64 dimensions. PCA and other fitted preprocessing are learned on training rows within each validation fold. The final serving transforms are fitted on development data, not test data.

4. Add patient and complaint features

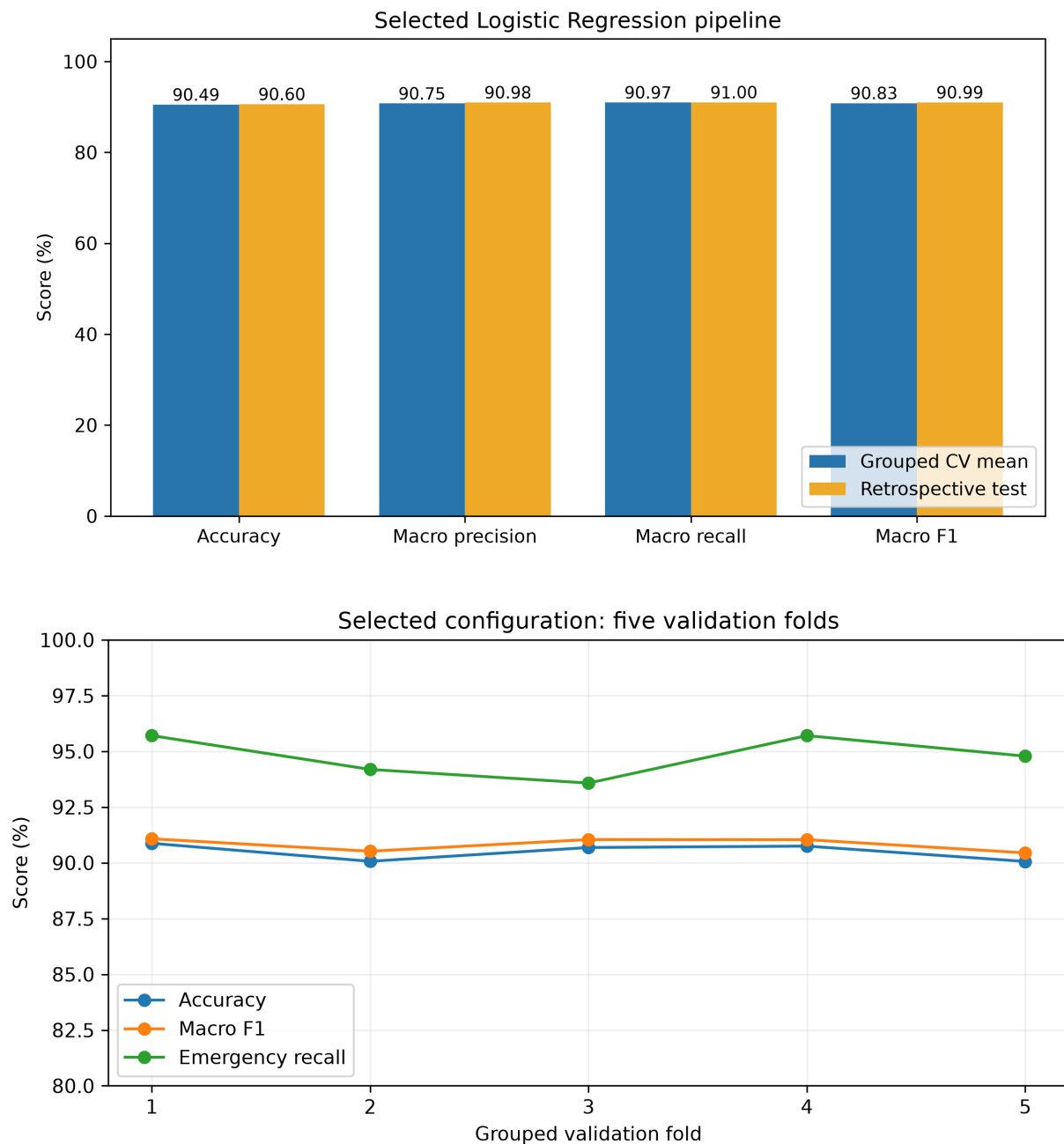
The 64 PCA components are combined with 50 patient features and 19 complaint-detail features, for 133 inputs. Patient features include age, heart rate, systolic and diastolic blood pressure, temperature, SpO2, AVPU, gender, arrival mode and ECG status. Numeric terms include quadratic combinations. Complaint details cover such items as severity, duration, onset and symptom mentions.

5. Predict one of four levels

One balanced Logistic Regression classifier with $C=100$ produces the class probabilities. The highest-probability class supplies the model prediction. Other evaluated classifier families are not used together in this deployed model.

The study uses supplied/recovered concepts; GUI translation can introduce a different error source. The report does not equate saved-concept accuracy with real-time translation performance.

Selected Model: Aggregate and Fold Scores

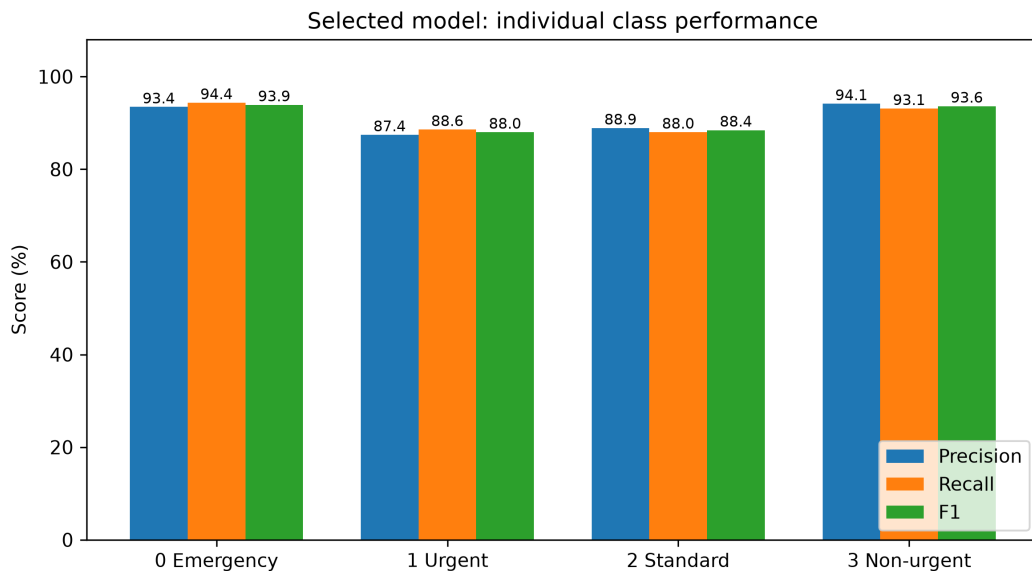
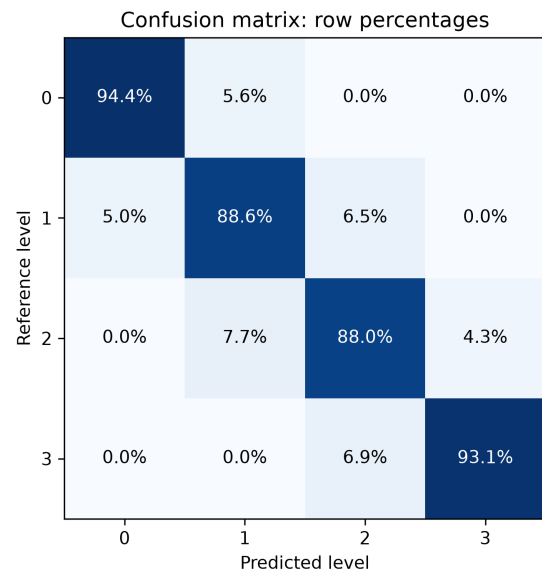
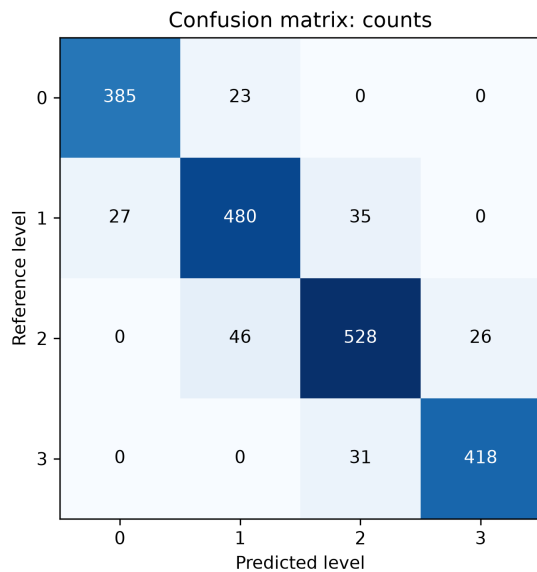


Grouped CV mean averages the five validation-fold scores. Related complaint groups stay together across fold boundaries. Retrospective test scores come from the fixed 1,999-row test partition. Both quantities are measured; neither guarantees performance on new clinical records.

This configuration was selected after a larger development search. Its CV scores are therefore subject to selection optimism. The previously examined test partition is not a fresh independent confirmation set.

Selected Model: Four-Level Predictions

Level	Precision	Recall	F1	Support
0 Emergency	93.45%	94.36%	93.90%	408
1 Urgent	87.43%	88.56%	87.99%	542
2 Standard	88.89%	88.00%	88.44%	600
3 Non-urgent	94.14%	93.10%	93.62%	449



Confusion-matrix rows are reference levels and columns are predictions. Counts total 1,999; row percentages divide by each reference class total. Urgent and Standard remain the most difficult classes.

Embedding Plots: Four Reproducible Samples

Current paired-text SapBERT embeddings by triage level
Four balanced samples; the same fitted PCA in every panel

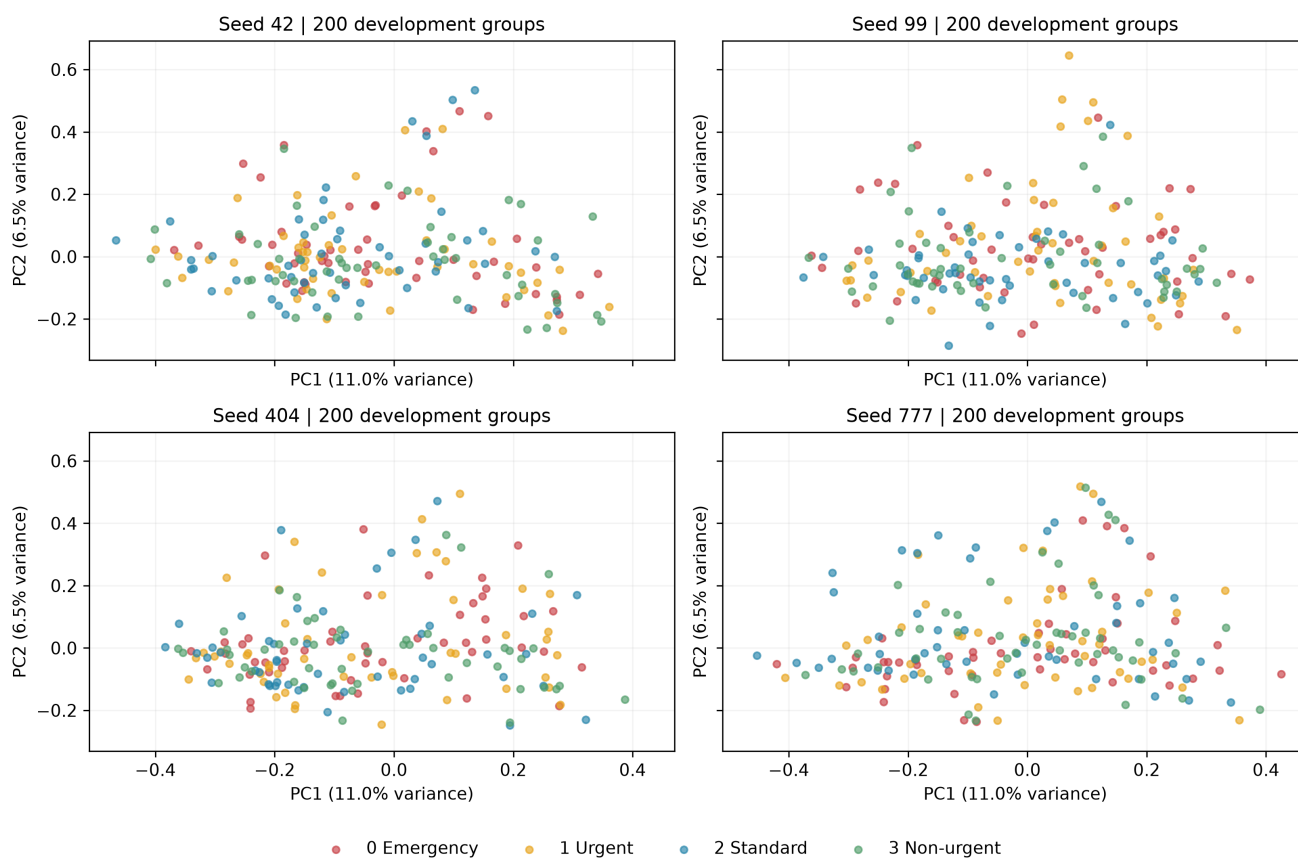


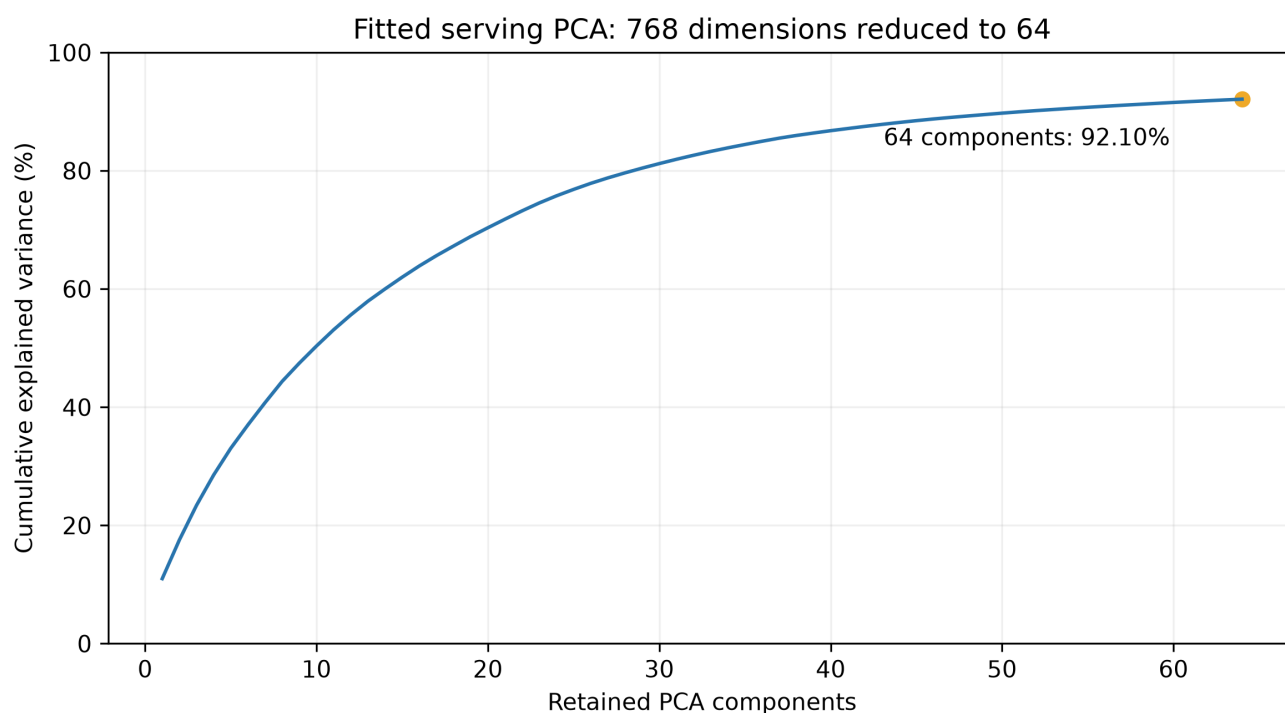
Figure 5. Two-dimensional PCA projections of current paired-text SapBERT embeddings. Each panel samples 200 distinct development complaint groups, balanced at 50 records per triage level. Seeds are 42, 99, 404 and 777. All panels use the same fitted serving PCA and shared axes. Colours indicate supplied triage levels 0-3; no test records are included.

Each point represents one record. Panels are repeated samples of the same dataset and can share records; they are not four classifiers, independent datasets or separately trained encoders. Group selection changes between panels, while the model and projection remain fixed.

Overlap does not contradict the classifier scores. The plot shows only two text components; the classifier uses 64 text components plus 69 patient and complaint-detail features. Similar complaints can have different urgency depending on patient measurements and other information.

These are triage-level plots, not the earlier arm-pain/back-pain category experiment. The current prepared dataset has no verified complaint-category column. No category labels or separated clusters were invented to reproduce that older appearance.

PCA and Embedding Geometry



The 64 selected components retain 92.10% of the embedding variance on the fitted development data. Explained variance measures retained embedding variation, not classification accuracy.

Space	Intra distance	Inter distance	Difference	Silhouette
768-D SapBERT	0.361 (0.006)	0.367 (0.006)	0.005 (0.002)	-0.019 (0.006)
64-D PCA	0.985 (0.005)	1.002 (0.003)	0.017 (0.006)	-0.011 (0.004)
2-D PCA display	0.991 (0.013)	0.997 (0.009)	0.006 (0.015)	-0.085 (0.016)

Table 3. Mean (sample standard deviation) over 20 reproducible development resamples, each with 200 distinct groups and 50 records per triage level. Intra/inter values are pairwise cosine distances within/between labels; difference is inter minus intra. Silhouette uses cosine distance. The same frozen embeddings and fitted PCA are reused. Samples may overlap; these are not 20 independent datasets or training runs.

Near-zero or negative silhouette indicates weak separation of triage labels in text geometry alone, not a performance ceiling for the combined classifier. Centering and projection change cosine geometry, so raw distances should not be treated as directly interchangeable across spaces. No ANOSIM, permutation-test significance or Cohen d is claimed. The accompanying CSV preserves all 60 sample-by-space measurements.

Verification and Additional Performance Measures

Measure	Selected model
Emergency recall	94.36%
Undertriage	4.20%
Overtriage	5.20%
Quadratic weighted kappa	0.9575
Mean absolute level error	0.0940
Matthews correlation coefficient	0.8737
Log loss	0.2104
Multiclass Brier score	0.1298

Undertriage means predicting a numerically higher, less urgent level than the supplied label. Overtriage means predicting a more urgent level. Kappa and MCC measure agreement; log loss and Brier score assess probabilities, with lower values preferred. These are not all percentage metrics.

Verification completed

The serving adapter reproduced all 1,999 selected predictions and probabilities. Thirteen live embedding checks and one live English prediction passed. All 49 targeted unit tests and the real six-tab GUI audit passed. Dataset identities, grouping boundaries and the fitted artifact hashes were checked. GUI, CLI and batch inference share the selected preprocessing and classifier contract.

The current report reads saved verified results; generating it does not retrain the model. Individual records and per-record predictions are excluded from the shared image package.

Why Accuracy Improved with the Same Model Family

The classifier family remains Logistic Regression and the SapBERT checkpoint remains frozen. However, the complete prediction pipeline changed: it now represents the supplied concept together with the original complaint, uses a 128-token input limit, retains 64 PCA components and uses balanced Logistic Regression with $C=100$. Patient-feature combinations and explicit complaint details are included.

The original complaint may preserve severity, timing or context that the supplied clinical concept omits. Giving the classifier this additional representation, while choosing suitable dimensionality and regularization, produced better measured aggregate scores. The experiments do not isolate exactly how much of the improvement comes from each simultaneous change.

What to tell your professor

“We retained the SapBERT encoder and Logistic Regression classifier family, but improved the input representation and tuned the prediction pipeline. SapBERT now encodes the clinical concept together with the original complaint. The selected model combines PCA-64 text features with patient measurements and complaint details. On the unchanged four-level dataset and grouped splits, it achieved 90.60% retrospective accuracy and 90.99% macro F1. The encoder was not fine-tuned, and the labels were not changed. Independent evaluation is still needed because the test records had been examined previously.”

Earlier uncertainty should be described accurately: 90% could not be guaranteed from the results available at that time. It was not established that improvement with this model family was impossible. The new measured result supports this particular pipeline, not a guarantee that every metric or every future dataset will exceed 90%.

This report includes only the selected deployed model. Full search records remain available separately for reproducibility; unsuccessful models are not replaced with invented high scores.

Semantic Concepts: Four Symptom-Based Runs

Method update | Read with the existing results on pages 1-8

The original manuscript grouped complaints by semantic concept, whereas the earlier figure on page 5 colours points by urgency. They answer different questions. Use the symptom-based figure below for the semantic analysis; keep the triage figure only as an auxiliary urgency visualization.

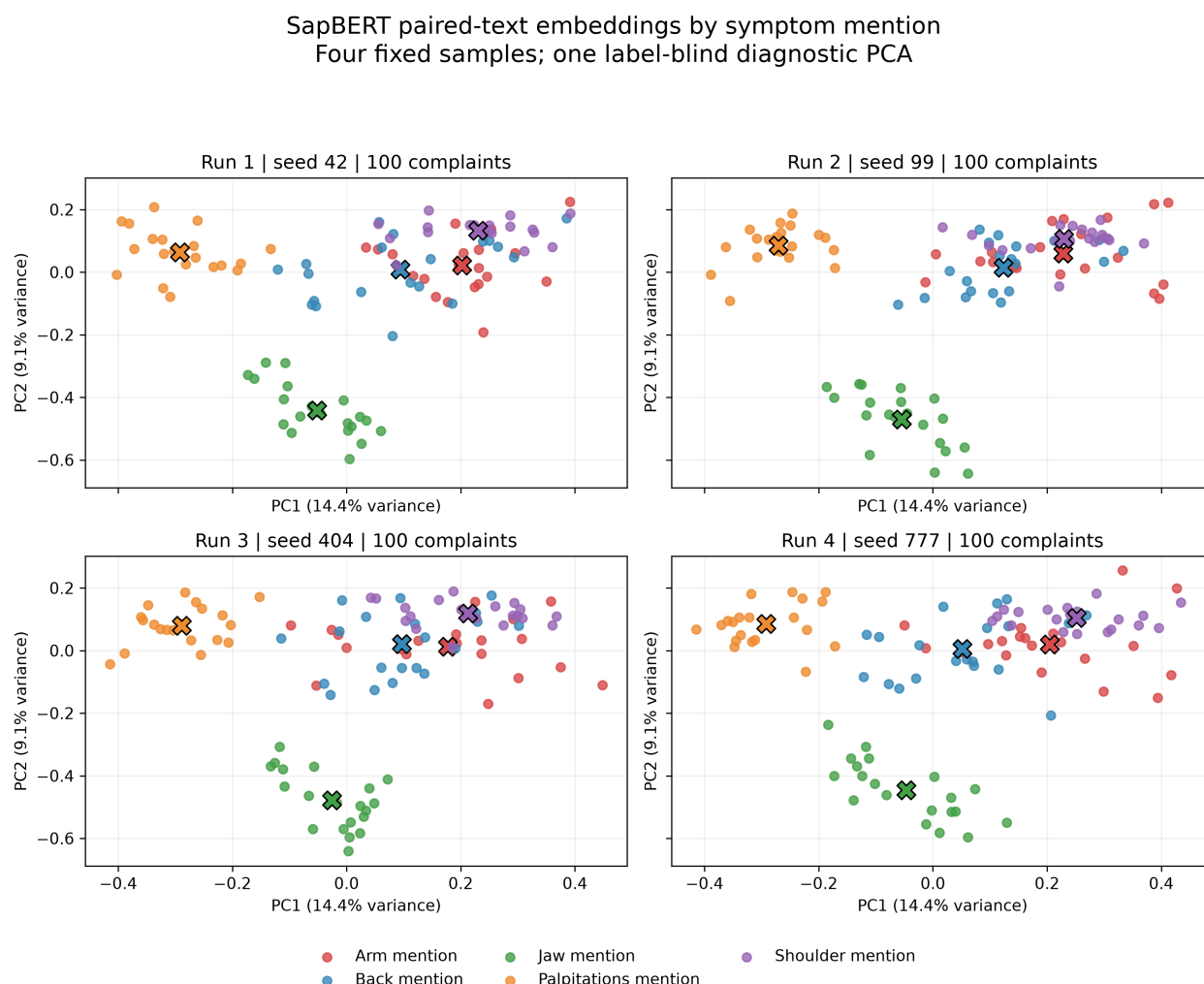


Figure 7. Paired-text SapBERT embeddings grouped by arm, back, jaw, palpitations and shoulder mentions. Each panel contains 100 development complaints, with 20 per group. Seeds: 42, 99, 404 and 777. Crosses indicate the arithmetic centre of each coloured group. One PCA is fitted without labels to all 1,915 eligible development groups and reused in every panel. It is a diagnostic projection, not a replacement for the deployed PCA-64 transform.

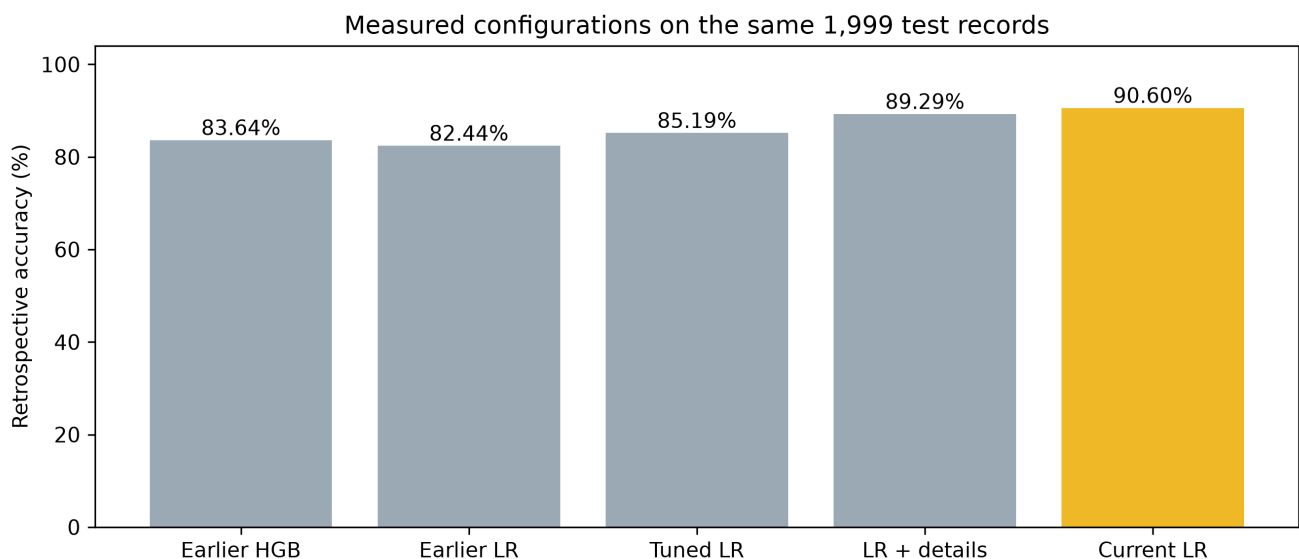
Groups are derived from literal mentions in Clinical_Concept; rows mentioning multiple listed groups are excluded. They are not clinician-verified diagnoses, so the labels say “mention” rather than assuming every arm/back/jaw/shoulder mention is a pain diagnosis. All four runs are retained regardless of appearance; samples can overlap between runs.

The old image used concept embeddings; these figures use the current concept-plus-complaint representation and different sampled records. Identical separation is not expected. Close points indicate similar projected text representations, not correct triage predictions. Overlap is reported rather than removed.

The Exact Change from About 83% to 90%

The saved four-level baseline closest to “83%” is full-768 HistGradientBoosting at 83.64%. The current selected classifier is Logistic Regression at 90.60%: a gain of 6.95 percentage points using unrounded scores. A same-family comparison starts with PCA-64 Logistic Regression at 82.44%, giving a gain of 8.15 percentage points.

Measured configuration	Accuracy	Precision*	Recall*	F1*
Earlier HGB / full 768	83.64%	84.72%	84.03%	84.21%
Earlier LR / PCA-64	82.44%	83.01%	83.22%	83.03%
Tuned LR / PCA-64	85.19%	85.71%	85.90%	85.80%
LR + quadratic + details / PCA-128	89.29%	89.67%	89.81%	89.73%
Current paired-text LR / PCA-64	90.60%	90.98%	91.00%	90.99%



*Precision, recall and F1 are macro averages across levels 0-3. Each row uses the same four-level data and previously examined 1,999-record partition. These are measured configurations, not isolated causal effects. The bars do not mean that each feature change alone caused the difference between neighbouring bars.

The tuned PCA-64 LR reference reaches 85.19%; the concept-only encoder with quadratic patient features, complaint details, PCA-128 and C=10 reaches 89.29%. The current paired-text/PCA-64/C=100 configuration reaches 90.60%. The final step adds 1.30 percentage points of accuracy and 1.26 points of macro F1, calculated before rounding.

The earlier and current encoders share the frozen SapBERT checkpoint. Changing the classifier family from the HGB reference must be acknowledged when quoting 83.64% to 90.60%; the same-family LR comparison is provided separately.

Method Improvements: What Was Actually Changed

1. Preserve more information in the encoder input

Previously SapBERT encoded the English clinical concept alone. The selected encoder input now combines the concept, [SEP] and original complaint. The token limit increased from 64 to 128 to accommodate the longer input. The checkpoint, CLS pooling and L2 normalization were retained; SapBERT was not fine-tuned.

2. Represent patient-feature combinations

The final patient block has 50 features: 35 linear, squared and pairwise numeric terms plus 15 categorical indicators. These terms let a linear classifier represent combinations of patient measurements. The earlier fixed fused baseline used 22 patient features. Patient measurements were already present in that baseline.

3. Preserve explicit complaint details

Nineteen features represent severity, onset, exertion, radiation, breathing, fainting, sweating, nausea, history, negation and duration-related clues. They supplement the embedding. Original complaints already supplied these separate features in the 89.29% intermediate model; adding the paired embedding did not introduce them for the first time.

4. Tune dimensions and classifier regularization

Candidate PCA sizes and classifier settings were evaluated on grouped development folds. The final choice is 64 text components and balanced Logistic Regression $C=100$, rather than the intermediate $\text{PCA-128}/C=10$. Larger C means weaker regularization. Final input: 64 text + 50 patient + 19 detail features = 133.

5. Select using development evidence

Five-fold grouped validation keeps related complaints in the same fold. Preprocessing is fitted inside each training fold. Selection maximizes mean macro F1 subject to the original emergency-recall constraint, with accuracy used to break ties. The final model is fitted on 8,001 development records.

6. Retain weighting and verify deployment

Balanced weighting adjusts each training record's contribution by class frequency; it does not create new records or change labels. It was already used in the LR baseline, so it cannot be claimed as a newly isolated cause of the gain. The shared GUI/CLI/batch adapter was checked against all 1,999 saved predictions and probabilities.

No new training was performed to draw these graphs or assemble this PDF. The deployed bundle is unchanged. Individual percentage gains cannot be assigned to each step without controlled ablation experiments.

Interpretation and Reproducible Evidence

Manuscript-ready method summary

We improved the prediction pipeline while retaining the frozen SapBERT encoder. The revised representation combines the English clinical concept with the original complaint, uses a 128-token input limit, and reduces the resulting 768-dimensional vector to 64 PCA components. These components are combined with quadratic patient features and 19 explicit complaint-detail features. Balanced Logistic Regression with $C=100$ was selected using five-fold grouped development validation. On the same previously examined four-level test partition, accuracy increased from 83.64% for the earlier full-768 HistGradientBoosting baseline to 90.60% for the selected pipeline. Compared with the earlier PCA-64 Logistic Regression baseline, accuracy increased from 82.44% to 90.60%. These comparisons reflect complete pipeline configurations, not the independent effect of any single change.

What the semantic plots establish

Symptom-concept colours describe similarity of text, whereas triage labels describe urgency. A similar complaint can receive different urgency labels because of patient measurements and clinical context. The diagnostic plot retains approximately 23.5% of embedding variance in two dimensions; its coordinates do not include the full classifier input. Visual separation therefore cannot be used as a substitute for accuracy, recall or F1.

Current scores and remaining errors

Current grouped CV means are 90.49% accuracy, 90.75% macro precision, 90.97% macro recall and 90.83% macro F1. Retrospective test values are 90.60%, 90.98%, 91.00% and 90.99%, respectively. Some Urgent and Standard class scores remain below 90%. From the 89.29% intermediate pipeline, Emergency recall decreases from 95.10% to 94.36% even though overall undertriage improves from 4.65% to 4.20%.

Scope of the evidence

Labels and group partitions were unchanged. The test partition had been examined during the development history, and the selected CV scores are also subject to model-selection optimism. The results are retrospective and are not independent clinical validation. Saved English concepts were used for the reported study; live Ollama translation performance remains a separate evaluation. No claim is made that all future scores will exceed 90%.

Saved evidence used for this update

Baseline metrics/configurations: `output/results/triage_four_level/results.json`.

Current and intermediate results: `triage_four_level_round7/retrospective_results.json` and `previous_round_result.json` under `output/results/`.

Selection and CV: `triage_four_level_round7/selection.json` and `cross_validation.csv`.

Semantic sampling and provenance: `reports/complaint_semantic_audit/protocol.json` and `verification.json`.

Individual sample identifiers remain local; the shared package contains figures, aggregate metrics and provenance only.

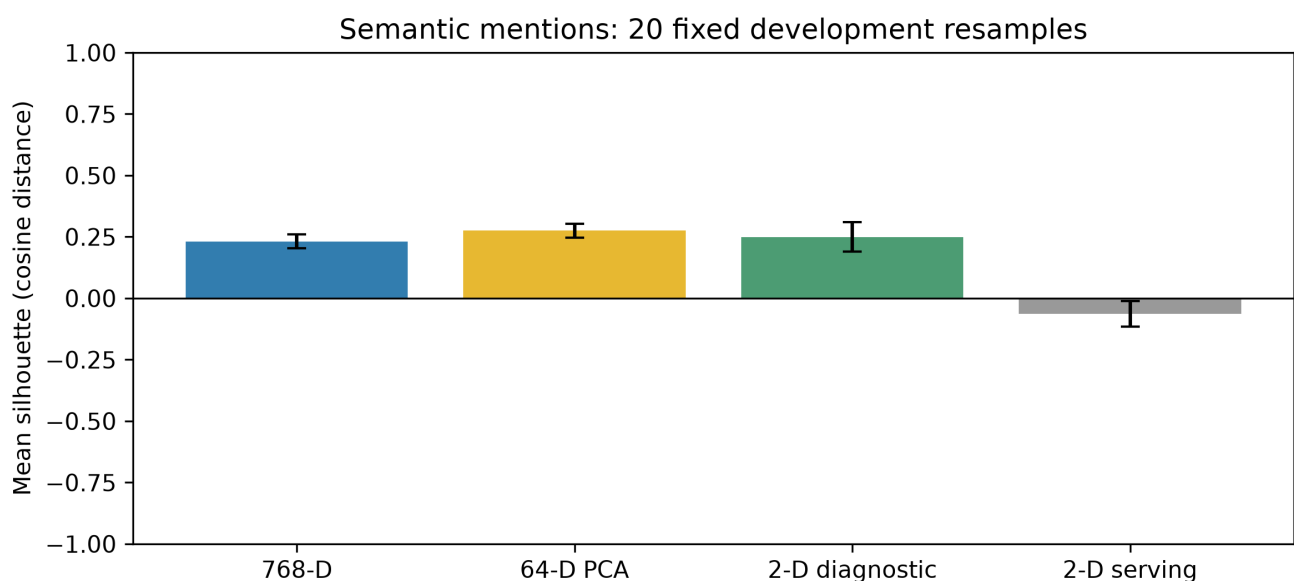
Semantic Separation: Actual Silhouette Values

Silhouette is bounded by -1 and +1. A value above +1 is invalid. Positive values indicate closer average similarity to the assigned group than to the nearest alternative group; values near zero indicate overlap. A negative value can occur when text-based distances do not agree with the chosen labels. It is not a classification accuracy percentage.

Table 4. Semantic-mention geometry. Twenty fixed samples of 50 development complaints (10 per mention group). Intra/inter values are means +/- sample SD across resamples; remaining columns are means. Colours/groups refer to arm, back, jaw, palpitations and shoulder mentions, not urgency.

Space	Intra	Inter	Diff.	d	R	Silhouette	Runs p<.05*
768-D	0.262 +/- 0.008	0.404 +/- 0.009	0.142	1.56	0.726	0.232	20/20
64-D PCA	0.636 +/- 0.021	1.015 +/- 0.011	0.379	2.04	0.786	0.275	20/20
2-D diagnostic	0.311 +/- 0.071	1.136 +/- 0.015	0.825	1.27	0.649	0.250	20/20
2-D serving	0.574 +/- 0.056	0.989 +/- 0.033	0.414	0.58	0.338	-0.064	20/20

*ANOSIM label-permutation p-values, Holm-adjusted across all 80 run/space tests. All values are recomputed from current paired-text embeddings. d is a descriptive standardized pair-distance difference and can exceed 1; silhouette cannot. R is ANOSIM rank separation. These are different statistics.



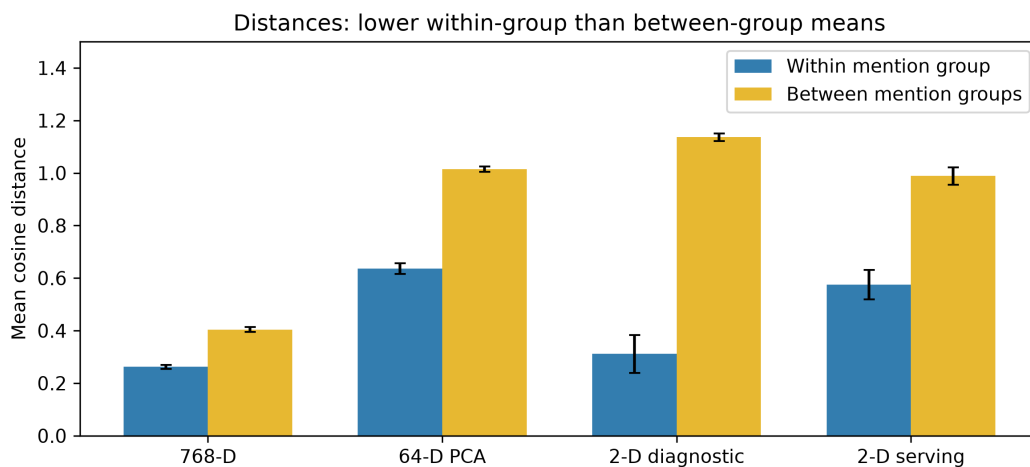
The four 100-record samples on page 9 previously gave mean silhouettes of 0.246 (768-D), 0.291 (64-D) and 0.289 (2-D diagnostic). This table uses 20 different 50-record samples under the repeated-sampling protocol; sampling explains the small numerical changes. No negative score was replaced with a positive one.

The 2-D serving view remains negative. The separate diagnostic projection, fitted without labels to the eligible development pool, answers a different display question. The deployed model and accuracy remain unchanged. Original triage-label geometry on page 6 remains an auxiliary result.

Within- and Between-Group Comparison

Table 5. Repeated-sample comparison. Each resample contributes one within-group mean and one between-group mean. Diff. = inter minus intra. A positive difference indicates larger average distances between groups. Samples overlap, so they are not 20 independent datasets.

Space	Mean diff.	SD diff.	Positive runs	Raw p range	Holm p range
768-D	0.142	0.010	20/20	0.0002-0.0002	0.016-0.016
64-D PCA	0.379	0.027	20/20	0.0002-0.0002	0.016-0.016
2-D diagnostic	0.825	0.064	20/20	0.0002-0.0002	0.016-0.016
2-D serving	0.414	0.059	20/20	0.0002-0.0004	0.016-0.016



Choice of statistical tests

Repeated random samples reuse development records, and pairwise distances within each sample share records. Treating these observations as independent would exaggerate statistical precision. The report therefore gives descriptive across-sample differences and per-sample record-label permutation tests, rather than presenting an across-run t-test or Wilcoxon result as independent clinical evidence.

Statistical protocol

Seeds 20261005-20261024 were fixed before computation; every run is retained. Each sample has 50 distinct complaint groups, drawn without replacement within the sample. ANOSIM uses 4,999 shuffled label assignments plus the observed arrangement, with $p = (\text{exceedances} + 1) / 5,000$. Holm correction covers 80 tests. Minimum resolvable raw p is 0.0002; no p=0 is reported.

Cohen d divides the inter-minus-intra distance difference by the pooled sample SD of pair distances. It is descriptive, with no independent-pair confidence interval. Cosine distances can range from 0 to 2; centering and projection alter their scale. Significant average separation can coexist with negative silhouette because silhouette compares each record with its nearest alternative group.

Mention labels come from the same text that is encoded. These tests describe lexical/semantic association, not an independently labelled semantic benchmark, clinical validity or triage accuracy. All fitted classifier artifacts and supplied labels are unchanged.

ED Triage Model Comparison

Table 6. Literature context. Scores are fractions (0.906 is 90.6%). Literature rows describe a different binary KTAS task and tenfold CV; our row is retrospective four-level evaluation. Metric definitions and inputs differ, so these rows do not establish superiority.

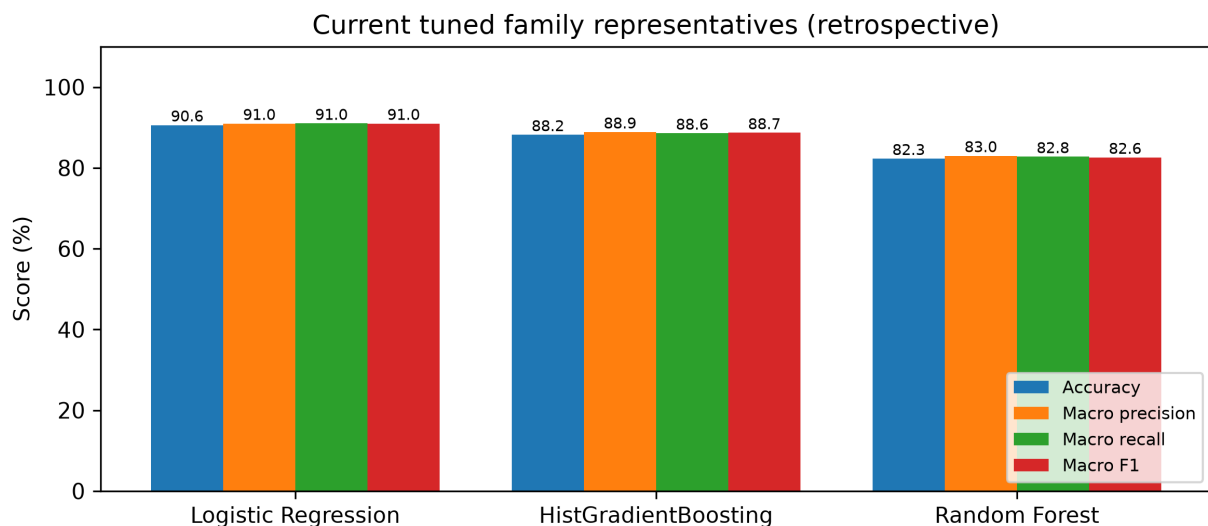
Model / evaluation	Accuracy	Recall	F1
TF-IDF + Logistic Regression [1]	0.750	0.988	0.544
TF-IDF + Random Forest [1]	0.751	0.964	0.582
BiLSTM [1]	0.746	0.846	0.670
Current paired SapBERT + PCA-64 + LR	0.906	0.910	0.910

[1] Seo et al. (2025), Artificial intelligence for severity triage based on conversations in an emergency department in Korea, Scientific Reports 15:16870, Table 2. DOI: 10.1038/s41598-025-99874-0. Published values are transcribed as reported, not recomputed. The 0.746/0.846/0.670 row is BiLSTM, not a combined BiLSTM-CNN-RNN hybrid.

The measured current four-level accuracy is 0.905953 (90.60%), rather than 0.990. The literature models above were not trained as part of our current comparison.

Table 7. Actual current three-family comparison. Same 1,999 test records and four-level targets. Precision, recall and F1 are macro averages. Each family uses its development-selected feature configuration; this is a pipeline comparison, not a controlled classifier-only ablation.

Family	Accuracy	Precision	Recall	F1
Logistic Regression	90.60%	90.98%	91.00%	90.99%
HistGradientBoosting	88.19%	88.86%	88.58%	88.68%
Random Forest	82.29%	83.00%	82.76%	82.58%



Only the selected Logistic Regression pipeline is deployed. The comparison retains HistGradientBoosting and Random Forest as measured alternatives. It does not imply that the GUI combines their predictions.

Classifier Matrices and Reproducibility

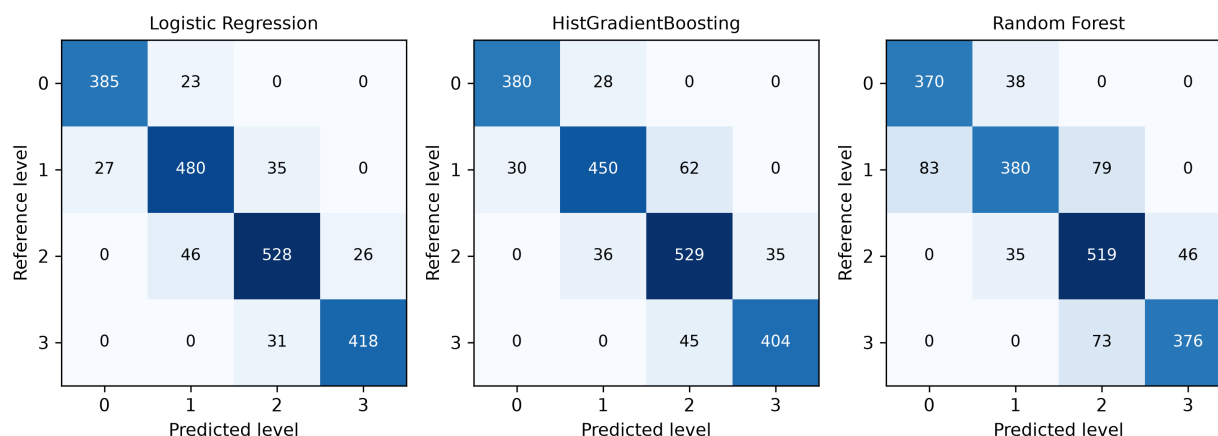


Figure 11. Current tuned family representatives. Rows are reference levels, columns are predicted levels, ordered 0 Emergency, 1 Urgent, 2 Standard and 3 Non-urgent. Each matrix contains 1,999 cases. Large diagonal counts indicate correct classifications; off-diagonal counts show errors.

How to read the complete report

Pages 1-4 contain the current selected model, aggregate metrics and class results. Pages 5-6 retain urgency-coloured text diagnostics as auxiliary evidence. Pages 7-8 provide model verification and input improvements. Pages 9-12 contain the symptom-based scatter plots and the measured progression from the earlier baseline to 90.60%. Pages 13-16 add semantic statistics, publication-style comparison tables and current three-family results.

What was recomputed in this update

Twenty fixed semantic resamples were measured in four embedding spaces, yielding 80 rows of geometry and permutation statistics. The model tables and confusion matrices were regenerated from saved verified experiment results. No classifier or encoder was retrained, and no label or plotted coordinate was altered to improve presentation.

Reproduction and sources

Semantic sampling and metric implementation: `experiments/triage_study/semantic_statistics.py`. Per-run aggregate results and protocol: `output/results/semantic_statistics/`.

Current family metrics and matrices: `output/results/triage_four_level_round7/family_results.json`. Selected configuration: `lr_pair64_c100`.

Figures and aggregate data are included in the current image ZIP. Private sampled records are excluded. The displayed mean $\pm$ SD describes resample variability, not uncertainty on an independent population estimate.

Silhouette definition: scikit-learn `silhouette_score` documentation. Literature source: Seo et al. (2025), Table 2, DOI 10.1038/s41598-025-99874-0.

The test partition has been examined previously; these scores remain retrospective. Neither positive semantic separation nor 90.60% retrospective accuracy establishes clinical validation or live translation performance.